

Bluestock Mutual Fund Analytics Capstone

Final Report | Data Analyst Internship Capstone | 2026

This report summarizes the end-to-end analytics solution developed for mutual fund industry, fund performance, investor behavior, SIP trends, and portfolio risk analysis. The project combines raw data ingestion, ETL, SQL warehousing, exploratory analytics, performance modeling, advanced risk metrics, and dashboard storytelling.

Deliverable	Status
Final report	Completed
12-slide presentation	Completed
Clean Python scripts	Completed
run_pipeline.py	Completed
README and checklist	Completed
Dashboard screenshots	Completed

1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone was designed to convert multiple mutual fund datasets into an analytical workflow that can support business reporting, fund comparison, investor segmentation, and risk assessment. The solution begins with raw CSV ingestion and live NAV fetches, then moves through cleaning, transformation, SQLite loading, notebook-based analysis, and Power BI dashboarding.

- The repository includes 16 raw CSV files, 22 processed analytical outputs, and a SQLite database for reproducible querying.
- The dashboard covers four business views: industry overview, fund performance, investor analytics, and SIP versus market trends.
- Performance analytics include returns, risk-adjusted ratios, alpha/beta, drawdown, tracking error, VaR/CVaR, SIP continuity, cohort analysis, and a risk-appetite recommender.
- The final package contains professional documentation and a master pipeline script so the project can be re-run from a clean environment.

Dashboard KPI Snapshot

Metric	Observed in dashboard
Total industry AUM	INR 81L Cr
Monthly SIP inflow	INR 31K Cr
Investor folios	26.12 Cr
Total schemes	1,908

2. Problem Statement and Objectives

Mutual fund stakeholders need a consolidated way to understand fund health, investor movement, category flows, and market-linked performance. Raw data alone is difficult to compare because the datasets use different grains: daily NAV, monthly SIP, fund-level performance, investor-level transactions, and portfolio holdings. The project objective was to create a single analytical submission that makes these views coherent and presentation-ready.

Objectives Covered

- Ingest all datasets and document data quality.
- Clean and standardize data types, date fields, transaction categories and numerical measures.
- Create a SQLite analytical warehouse using a star-schema style design.
- Produce EDA visuals for NAV, AUM, SIP, category inflows, investor demographics and folio growth.
- Calculate performance metrics for fund comparison and risk analysis.
- Build dashboard pages for business consumption.
- Prepare final report, presentation, README and clean code package.

3. Data Sources

The capstone uses ten structured datasets plus live NAV histories fetched from the public mfapi.in endpoint. The datasets collectively represent fund metadata, NAV history, industry AUM, SIP inflows, category inflows, folios, scheme performance, investor transactions, portfolio holdings and benchmark indices.

Dataset	Analytical Role
Fund master	Scheme metadata, AMC, category and expense attributes
NAV history	Daily trend, returns and correlation analysis
AUM by fund house	AMC scale and industry growth analysis
Monthly SIP inflows	SIP inflow and market-movement trend analysis
Category inflows	Category-wise net flow comparison
Industry folio count	Retail participation and folio growth
Scheme performance	Returns, alpha, beta, Sharpe, Sortino and drawdown
Investor transactions	Investor behavior by state, age group and transaction type
Portfolio holdings	Sector allocation and concentration risk
Benchmark indices	Benchmark comparison and beta calculations

4. ETL Design

The ETL pipeline follows a staged pattern: raw files are profiled, cleaned with pandas, written to processed CSV files, and loaded into SQLite. The SQLite layer acts as a lightweight analytical warehouse for repeatable SQL queries and dashboard connectivity.

Layer	Output
Raw ingestion	CSV files in data/raw and profiling report
Cleaning	Standardized processed CSV files
Transformation	Derived date dimension, daily returns and quality flags
Storage	SQLite database: data/db/bluestock_mf.db
Consumption	Notebooks, SQL queries and Power BI dashboard

Cleaned Code Components

- data_ingestion.py profiles raw files and saves a compact summary.
- live_nav_fetch.py fetches selected live NAV histories.
- day2_clean_load.py runs cleaning and database loading.
- run_pipeline.py acts as the master execution script.
- recommender.py returns top funds by risk appetite and Sharpe ratio.

5. Database and Analytical Model

The analytical model is organized around fund and date dimensions with fact tables for NAV, transactions, performance, AUM, SIP industry trend, category inflows, folios, portfolio holdings and benchmark indices. This structure makes the dashboard and SQL analysis easier to maintain because each dataset keeps its natural grain while sharing common keys where possible.

- dim_fund stores fund metadata.
- dim_date standardizes calendar attributes.
- fact_nav stores daily NAV and daily return percentages.
- fact_transactions stores investor-level transaction facts.
- fact_performance stores fund-level performance and risk measures.
- Supporting fact tables store industry, category, portfolio and benchmark measures.

Data Volume Snapshot

Item	Count
Processed files	22
Funds / schemes	40
Fund houses	10
NAV rows	64,320
Transaction rows	32,778

6. EDA Findings: NAV, AUM and SIP Trend

NAV and AUM analysis indicates how fund performance and AMC scale evolved across the study period. SIP inflow analysis adds the investor participation angle and links recurring investment behavior with broader market movement.

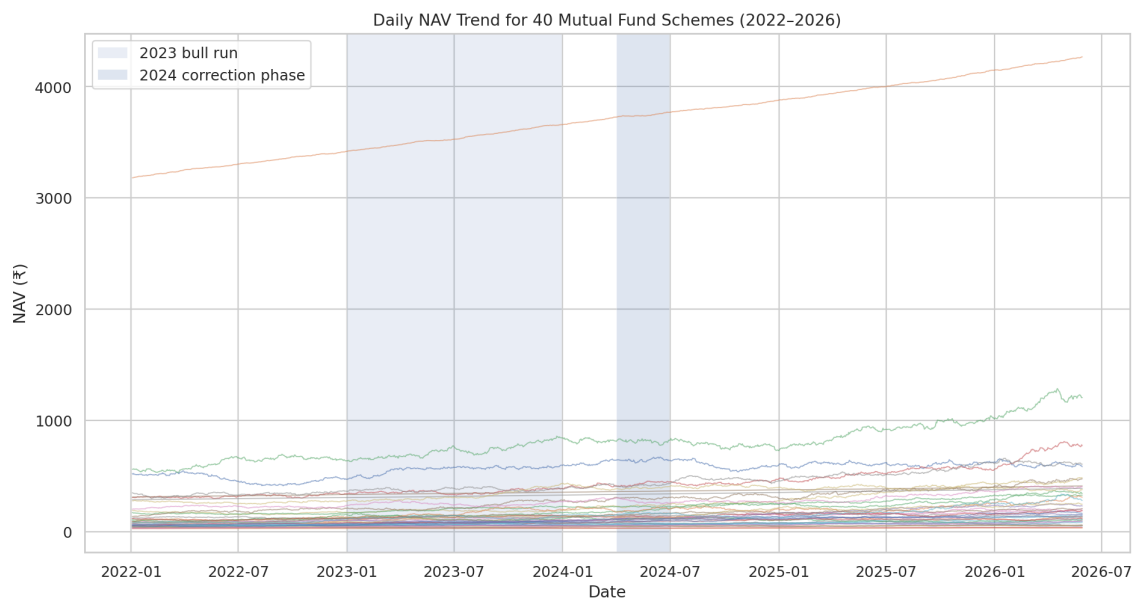


Figure 1: NAV trend across schemes.

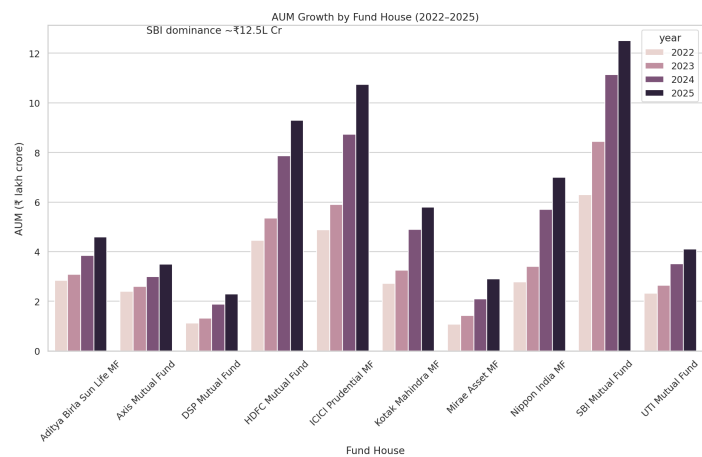


Figure 2: AUM growth by fund house.

7. EDA Findings: Category, Demographic and Folio Analysis

The investor analytics view shows how transaction behavior changes by age group, geography and transaction type. Category inflow analysis highlights which mutual fund segments attracted higher net flows in the most recent period.

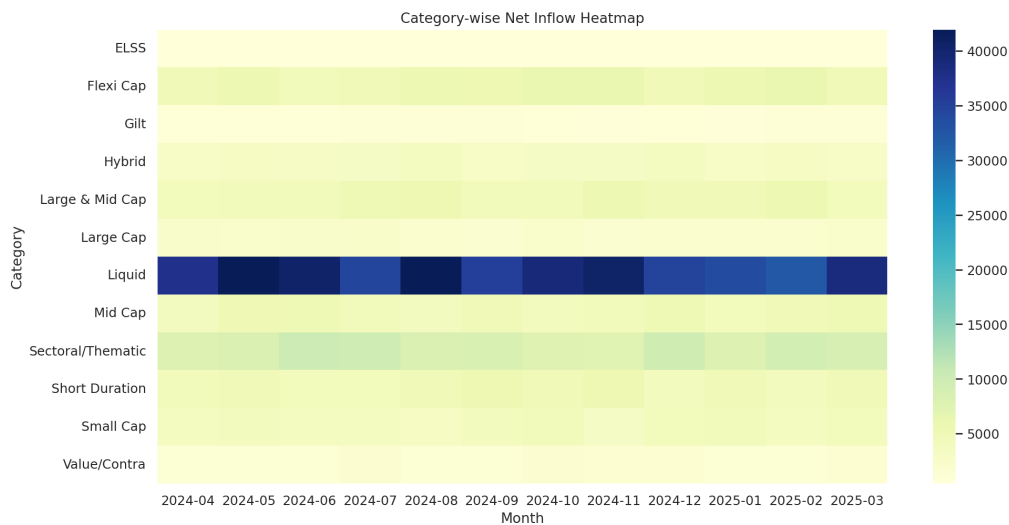
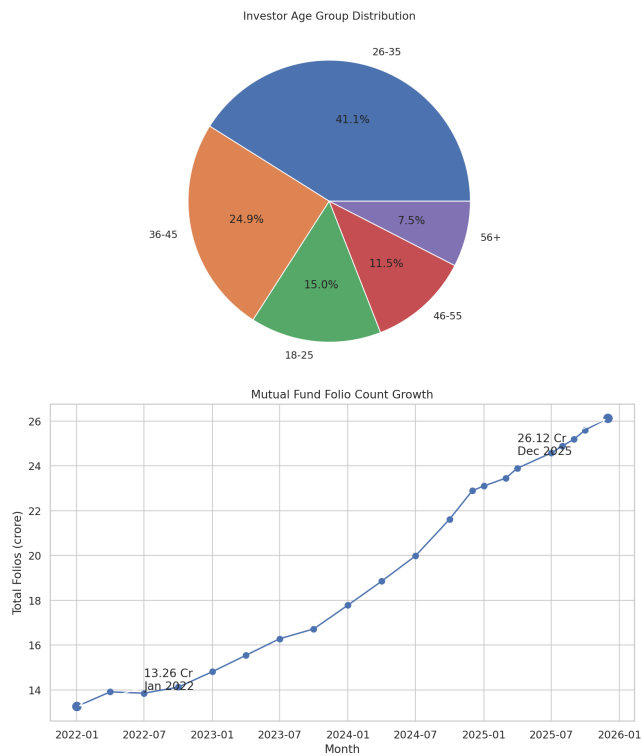


Figure 3: Category-wise monthly net inflows.



8. Performance Analysis Methodology

Fund performance was evaluated using both absolute and risk-adjusted measures. Absolute return metrics show growth, while risk metrics indicate whether the return was achieved efficiently and consistently. This is important because a high-return fund may still be unsuitable if drawdown, volatility or downside risk is high.

Metric	Purpose
CAGR	Annualized long-term return comparison
Sharpe Ratio	Excess return per unit of total volatility
Sortino Ratio	Return efficiency after penalizing downside volatility
Alpha	Excess return versus benchmark expectation
Beta	Sensitivity to market benchmark movement
Max Drawdown	Worst peak-to-trough decline
Tracking Error	Deviation from benchmark path
VaR / CVaR	Potential downside loss under historical risk scenarios

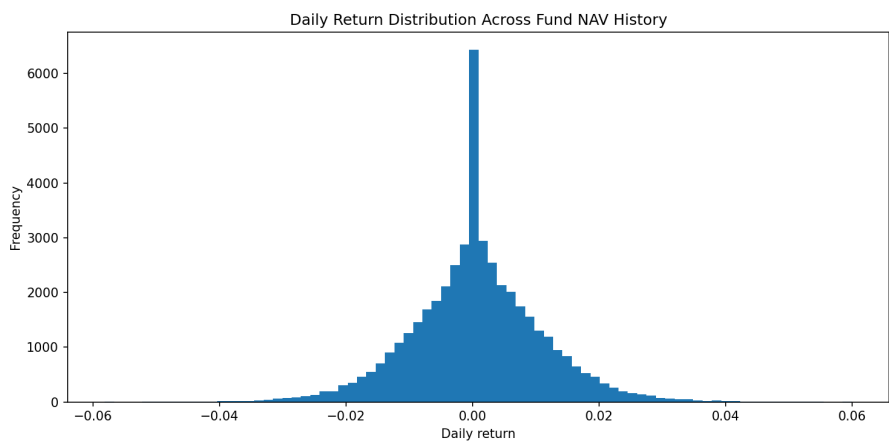


Figure 4: Daily return distribution.

9. Performance Results and Scorecard

The fund scorecard ranks schemes by a composite view of return, volatility, Sharpe ratio, expenses and drawdown. This helps compare schemes beyond a single return number and supports dashboard filtering by fund house or category.

scheme_name	return_3yr_pct	sharpe_ratio	expense_ratio_pct	max_drawdown_pct
Mirae Asset Large Cap Fund -	33.99	1.73	1.46	-11.27
ICICI Pru Midcap Fund - Regu	31.77	1.42	1.36	-18.19
Kotak Flexicap Fund - Regula	29.58	1.56	1.45	-12.97
HDFC Mid-Cap Opportunities F	32.43	1.31	1.38	-16.22
ICICI Pru Bluechip Fund - Di	32.48	1.19	0.8	-12.59
Axis Midcap Fund - Regular -	35.1	1.19	1.38	-20.96
SBI Bluechip Fund - Regular	30.45	1.42	1.54	-15.01
Mirae Asset Tax Saver Fund -	29.17	1.48	1.6	-16.4

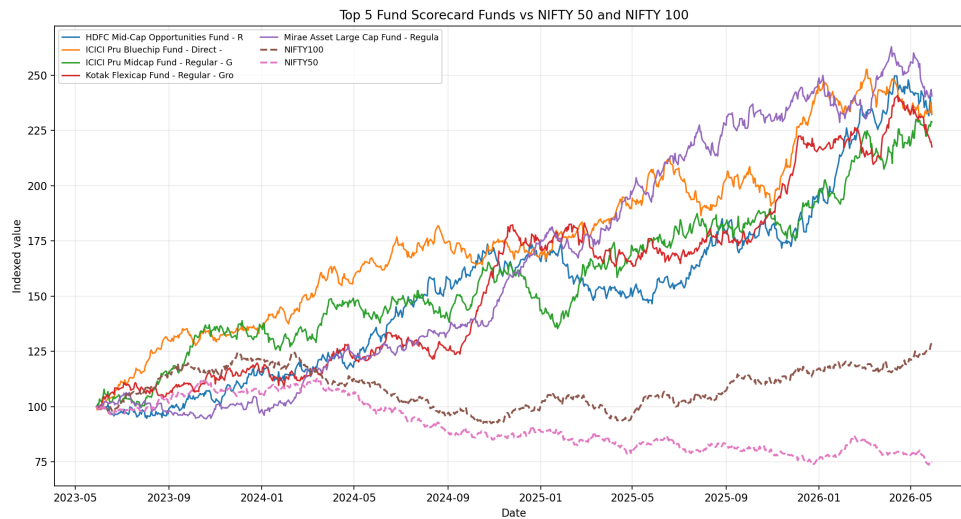


Figure 5: Top fund benchmark comparison.

10. Advanced Analytics

Advanced analytics extended the project from descriptive reporting into decision support. The work includes VaR and CVaR risk estimation, rolling Sharpe stability, SIP continuity checks, investor cohort analysis, sector concentration via HHI, and a simple risk-appetite recommender.

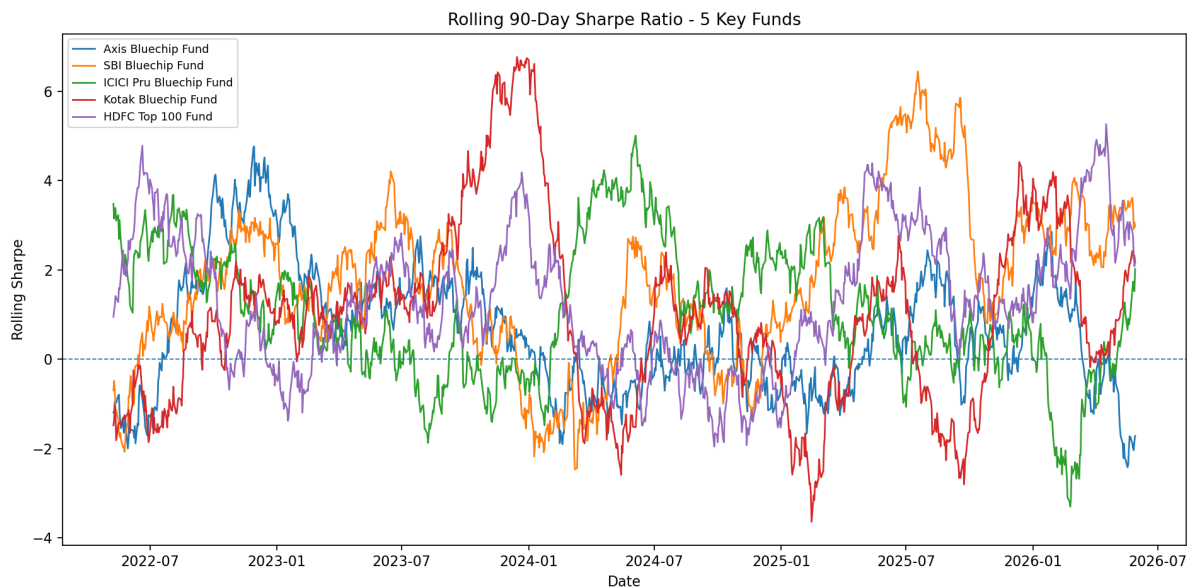
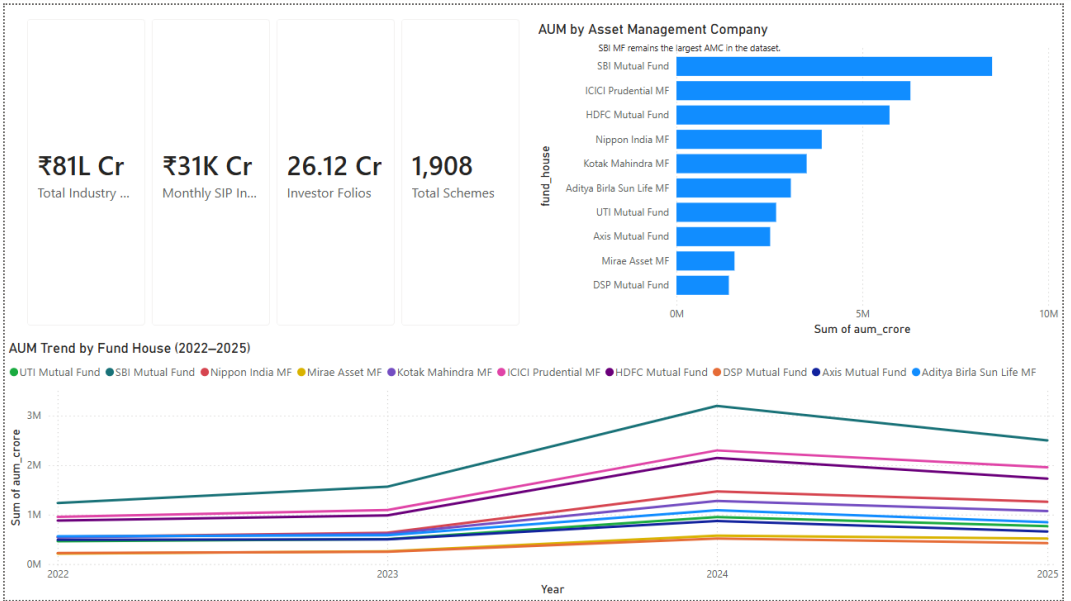


Figure 6: Rolling 90-day Sharpe ratio trend.

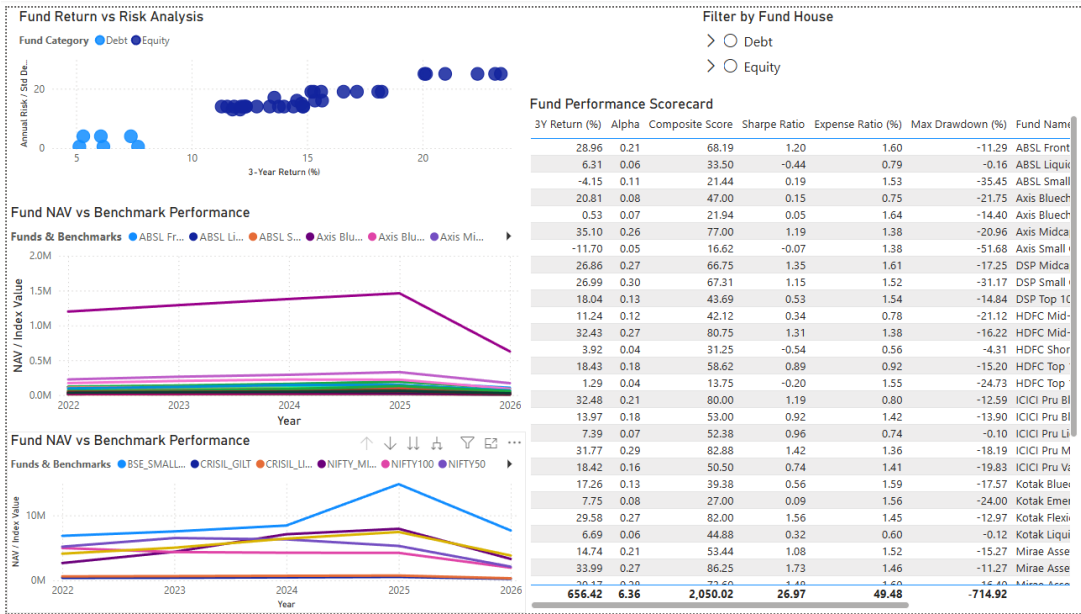
- Historical VaR and CVaR estimate downside risk from the actual return distribution.
- SIP continuity identifies possible investor drop-off behavior when gaps exceed expected SIP intervals.
- Sector HHI provides a portfolio concentration lens for diversification analysis.
- The recommender filters available funds by risk appetite and ranks them using Sharpe ratio.

11. Dashboard Screenshots: Industry and Fund Performance

The dashboard translates the analytical work into four business-ready pages. Page 1 provides total industry AUM, monthly SIP inflow, folios, total schemes, AMC trend and AMC ranking. Page 2 shows fund return-risk scatter, scorecard table and NAV versus benchmark analysis.



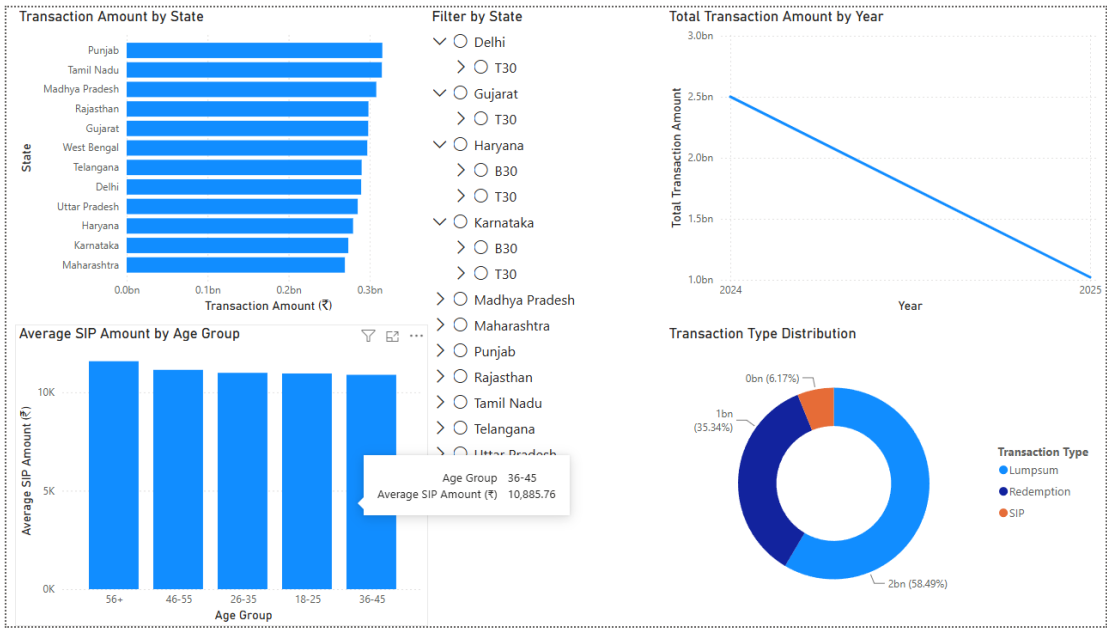
Dashboard Page 1: Industry Overview.



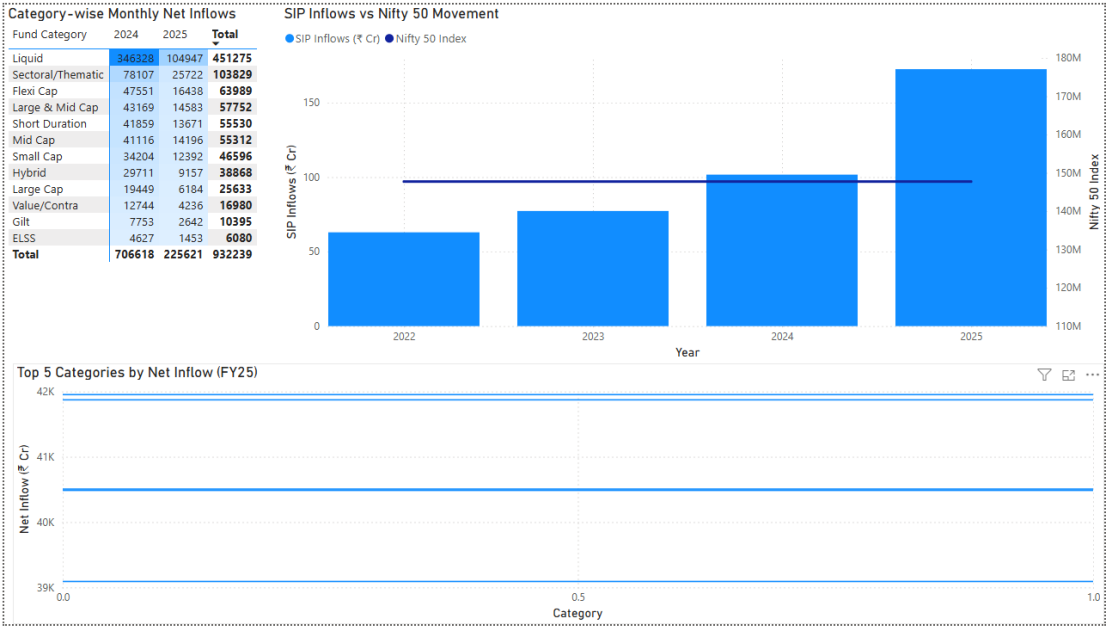
Dashboard Page 2: Fund Performance.

12. Dashboard Screenshots: Investors and SIP Trends

Page 3 focuses on investor transactions by state, transaction type, age group and time. Page 4 combines SIP inflows, Nifty movement, category-wise monthly inflows and top categories by FY25 net inflow.



Dashboard Page 3: Investor Analytics.



Dashboard Page 4: SIP and Market Trends.

13. Key Findings

- SBI Mutual Fund appears as the largest AMC in the dashboard dataset, with ICICI Prudential, HDFC, Nippon India and Kotak also contributing major AUM.
- SIP inflows and folio metrics show strong retail participation, making SIP trend monitoring an important industry KPI.
- Equity and debt funds occupy different regions of the return-risk view, confirming that fund category should be considered before ranking schemes.
- Investor transaction analytics show meaningful geographic and demographic segmentation opportunities.
- Fund scorecards are more useful than simple return ranking because they combine return, risk, expenses and downside behavior.

Business Value

The final dashboard helps analysts and stakeholders identify high-performing funds, monitor industry direction, understand investor behavior, and communicate insights using a professional visual layer. The reproducible pipeline ensures the analysis can be refreshed when new data arrives.

14. Limitations

- The analysis depends on the quality and completeness of supplied CSV files.
- Live NAV fetching requires network connectivity and mfapi.in availability.
- Some dashboard metrics are based on sample or capstone data and should be validated against official AMFI/AMC sources before production use.
- Risk metrics use historical behavior and cannot guarantee future performance.
- Power BI dashboard publishing was optional; the submitted package includes PDF and image exports for offline review.
- SQLite is suitable for capstone scale but a cloud database or warehouse would be preferred for production refresh and access control.

15. Recommendations

- Add an automated data validation layer using tests for missing values, duplicate keys, negative NAV values and invalid transaction categories.
- Schedule periodic NAV and benchmark refreshes to keep performance metrics current.
- Publish the dashboard to Power BI Service or Tableau Public and add the URL to README after publishing.
- Add fund-category filters to the recommender so equity, debt and hybrid funds are compared separately.
- Store data-refresh logs to track pipeline health over time.
- Extend the dashboard with drill-through pages for individual fund profiles and investor segments.

Submission Summary

All Day 7 deliverables are included in the final package: Final_Report.pdf, Bluestock_MF_Presentation.pptx, clean scripts, run_pipeline.py, README.md, dashboard screenshots, Git command notes and self-review checklist.

End of Report